The Amazon Relational Database Service (Amazon RDS) offers ready-to-use relational databases such as PostgreSQL, MySQL, MariaDB, Oracle Database, and Microsoft SQL Server. Beyond that, AWS offers its own engine called Amazon Aurora, which is MySQL and PostgreSQL compatible. As long as your application uses one of these database systems, it is not a big deal to switch to RDS. The trickiest part is migrating the data, which you will learn about in this chapter as well.

RDS is a managed service. The managed service provider—in this case, AWS—is responsible for providing a defined set of services—in this case, operating a relational database system. Table 10.1 compares using an RDS database and hosting a database yourself on virtual machines.

Table 10.1 Managed service RDS vs. a self-hosted database on virtual machines

	Amazon RDS	Self-hosted on virtual machines
Cost for AWS services	Higher because RDS costs more than virtual machines (EC2)	Lower because virtual machines (EC2) are cheaper than RDS
Total cost of ownership	Lower because operating costs are split among many customers	Much higher because you need your own manpower to manage your database
Quality	AWS professionals are responsible for the managed service.	You'll need to build a team of professionals and implement quality control yourself.
Flexibility	High, because you can choose a relational database system and most of the configuration parameters	Higher, because you can control every part of the relational database system you installed on virtual machines